Winter flounder (*Pseudopleuronectes americanus*)

Summary Scores

- **Sensitivity:** Moderate - High
- **Exposure:** Moderate - High
- **Vulnerability:** Moderate - High

Sensitivity

Sensitivity Attributes	Range			Gulf of Maine			Georges Bank			Southern New England/Mid-Atlantic		
	Score	LH DQ	Tally	Score	LH DQ	Tally	Score	LH DQ	Tally	Score	LH DQ	Tally
Adult Mobility	1.67	2.8		1.64	2.8		1.6	2.8		1.76	2.8	
Complexity in Reproductive Strategy	3.43	2.64		3.44	2.55		3.32	2.68		3.52	2.68	
Dispersal of Early Life Stages	2.37	2.8		2.44	2.7		2.28	2.8		2.4	2.9	
Egg and Larval Survival and Settlement Requirements	2.92	2.75		2.92	2.74		2.68	2.75		3.16	2.75	
Historical Variability in Temperature	2.03	2.87		2.2	2.9		1.96	2.8		1.92	2.9	
Juvenile Habitat Requirements	2.23	2.9		2.32	2.9		2.04	2.9		2.32	2.9	
Other Stressors	2.85	2.68		2.92	2.55		2.56	2.75		3.08	2.75	
Population Growth Rate	1.91	2.81		2	2.68		1.92	2.88		1.8	2.88	
Prey Specificity	1.57	3		1.56	3		1.52	3		1.64	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.24	2.9		1.24	2.9		1.16	2.9		1.32	2.9	
Spawning Cycle	2.84	2.77		2.84	2.7		2.84	2.8		2.84	2.8	
Stock Size/Status	2.65	2.53		2.72	1.9		2.96	2.9		2.28	2.8	
Total Sensitivity | Certainty	**Moderate (2) | 0.81**			**Moderate (2) | 0.58**			**Moderate (2) | 0.58**			**High (3) | 0.91**		

Exposure

Exposure Factors	Range			Gulf of Maine			Georges Bank			Southern New England/Mid-Atlantic		
	Mean	SD	Tally	Mean	SD	Tally	Mean	SD	Tally	Mean	SD	Tally
Bottom Temperature	2.92	0.98		2.65	1.07		2.7	0.86		3.42	0.65	
Bottom Oxygen	3.15	0.82		3.24	0.78		2.89	0.71		3.39	0.76	
Bottom Salinity	2.76	0.89		2.09	0.89		3.5	0.63		2.83	0.61	
Bottom Aragonite Solubility	1.05	0.26		1.06	0.27		1.08	0.35		1	0.04	
Net primary production	1.01	0.03		1.01	0.03		1	0.00		1.01	0.04	
Downward Flux of Particulate Organic Carbon	1.21	0.26		1.24	0.32		1.27	0.24		1.12	0.19	
Total Exposure	**Moderate (2.37)**	**0.90**		**Moderate (2.21)**	**0.89**		**Moderate (2.29)**	**0.98**		**High (2.66)**	**0.72**	

Vulnerability

Vulnerability	Range			Gulf of Maine			Georges Bank			Southern New England/Mid-Atlantic		
	Mean	SD	Tally	Mean	SD	Tally	Mean	SD	Tally	Mean	SD	Tally
Total Vulnerability	**Moderate (5.84)**	**2.71**		**Moderate (4.41)**	**1.79**		**Moderate (4.58)**	**1.95**		**High (7.93)**	**2.19**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Winter flounder. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 6 and Figure 5). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High	Moderate - High	Decrease - None
Exposure	Very High	Moderate - High	Decrease
Vulnerability	Very High	Moderate - High	Decrease
Directional Effect	Negative	Negative	None

Stocks

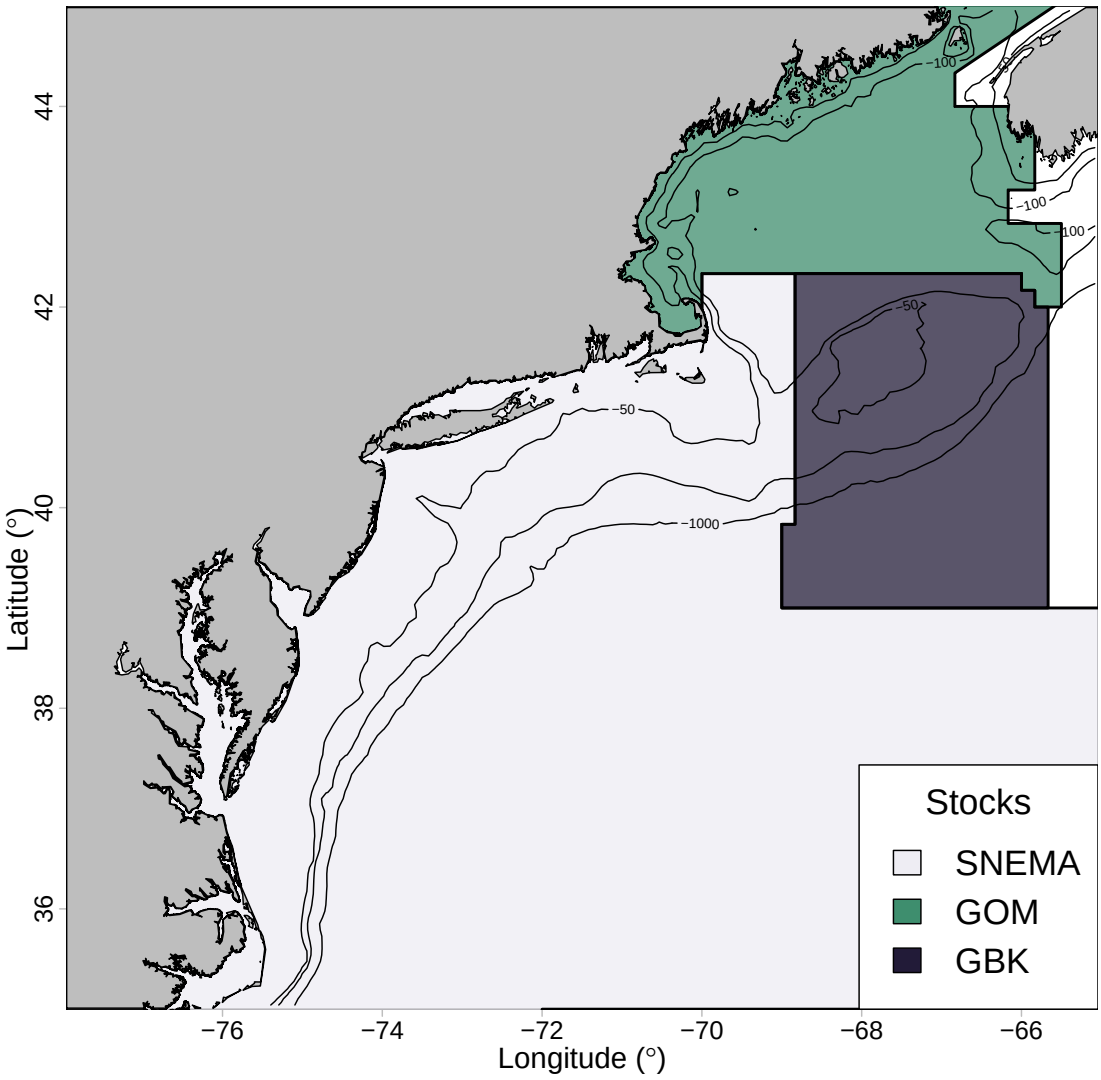


Figure 2: Winter flounder Stocks along the Northeast Shelf

Score Descriptions:

Sensitivity:

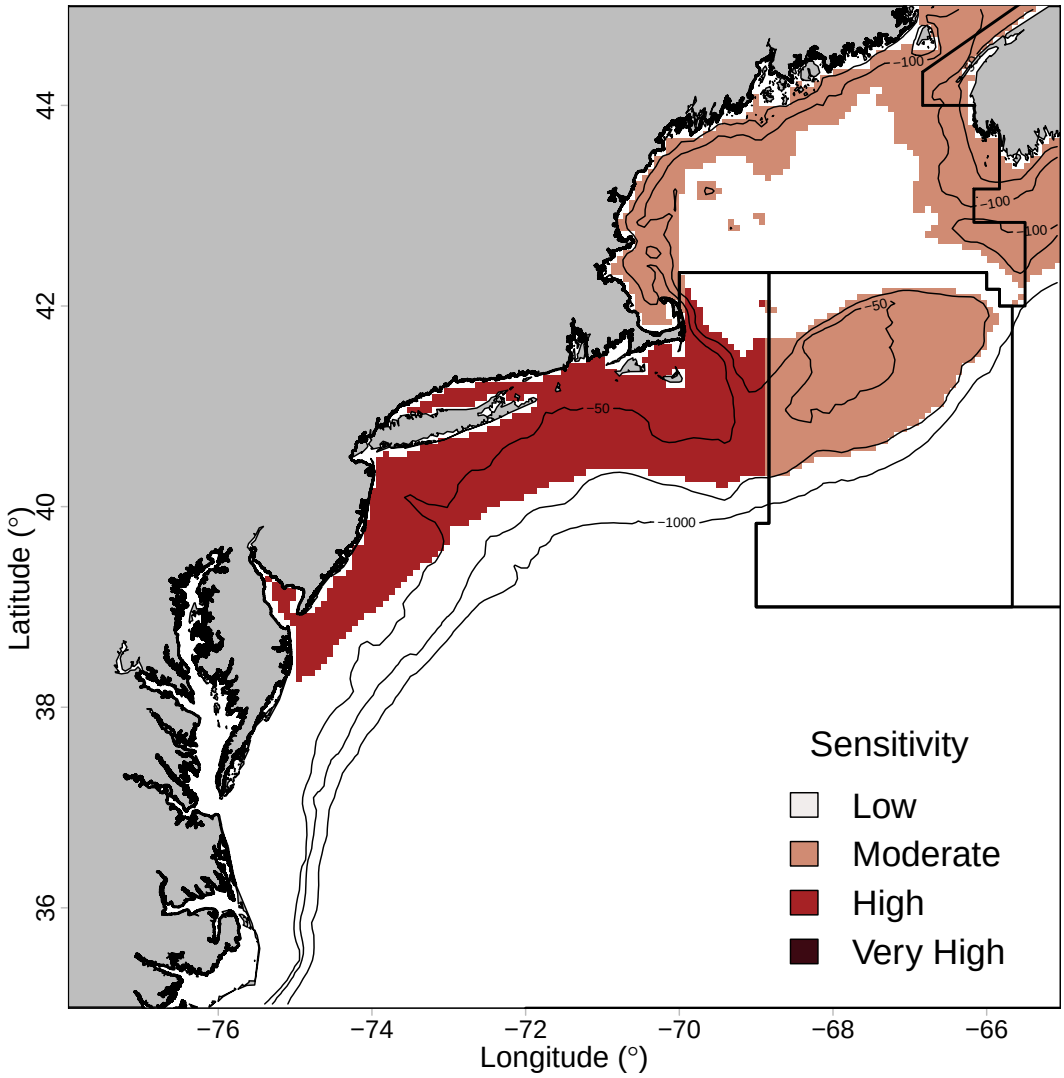


Figure 3: Winter flounder Sensitivity on the Northeast Shelf

The **Georges Bank** and **Gulf of Maine** stocks of Winter Flounder both received **Moderate** sensitivity scores. Certainty for both stocks was 58%. For these stocks, five attributes had scores greater than 2.5: Complexity in Reproductive Strategy (3.3 and 3.4), Egg and Larval Survival and Settlement Requirements (2.7 and 2.9), Other Stressors (2.6 and 2.9), Spawning Cycle (2.8), and Stock Size (3 and 2.7) (Figure 1).

The **Southern New England/Mid-Atlantic** stock received a **High** sensitivity score, with a certainty of 91%. Three attributes had scores greater than 3: Complexity in Reproductive Strategy (3.5), Egg and Larval Survival and Settlement Requirements (3.2), and Other Stressors (3.1) (Figure 1).

Winter flounder spawn in winter and early spring before waters reach temperatures of 6C (Pereira 1999). Spawning occurs in nearshore and estuarine habitats, including muddy and sandy substrates, and native salt marsh habitats which are considered highly vulnerable (Guyant et al. 2022). Larvae are dependent on

bottom currents to help retain them in and around spawning areas and nursery habitats. Adults depend on estuarine habitats, which may be impacted by hypoxia and changes to freshwater inputs (Guyant et al. 2022). Predators include striped bass, spiny dogfish, and seals, which have all experienced distribution shifts that could result in increased predation pressure (Pereira 1999).

These results are similar to those from the 2016 Climate Vulnerability Assessment where winter flounder were given a High sensitivity score. This score was also driven by Stock Status and Early Life History Requirements (Hare et al. 2016).

Exposure:

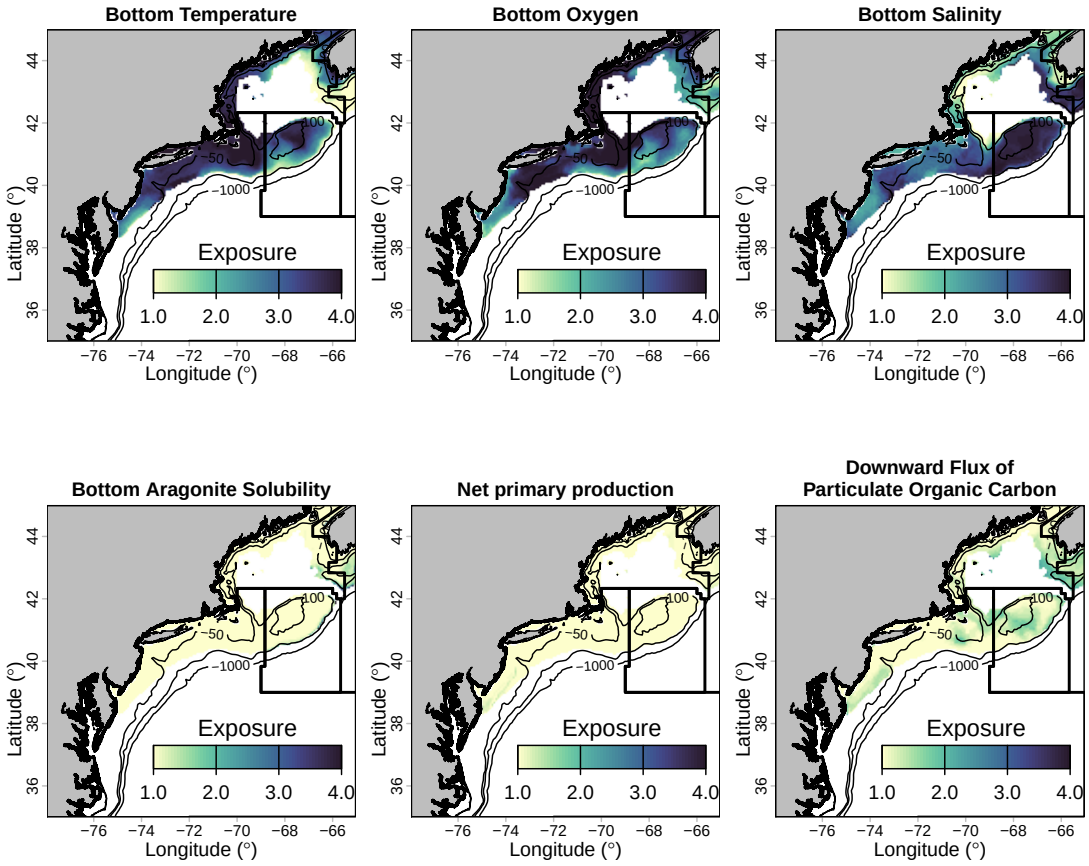


Figure 4: Maps of each Exposure Factor for Winter flounder. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure throughout the range of Winter Flounder was **Moderate** (2.37 +/- 0.90) (Figure 1). Within the **Gulf of Maine** and **Georges Bank** stocks mean exposure was **Moderate** (2.21 +/- 0.89 and 2.29 +/- 0.98, respectively). In the **Southern New England/Mid Atlantic** stock, mean exposure was **High** (2.66 +/- 0.72) (Figure 1). These patterns were driven by bottom temperature, oxygen, and salinity exposure (Figure 5, Figure 4).

In the previous Assessment, winter flounder received a Very High exposure score (Hare et al. 2016). This previous score was driven by high exposure to sea surface temperature, air temperature, and ocean acidification (Hare et al. 2016).

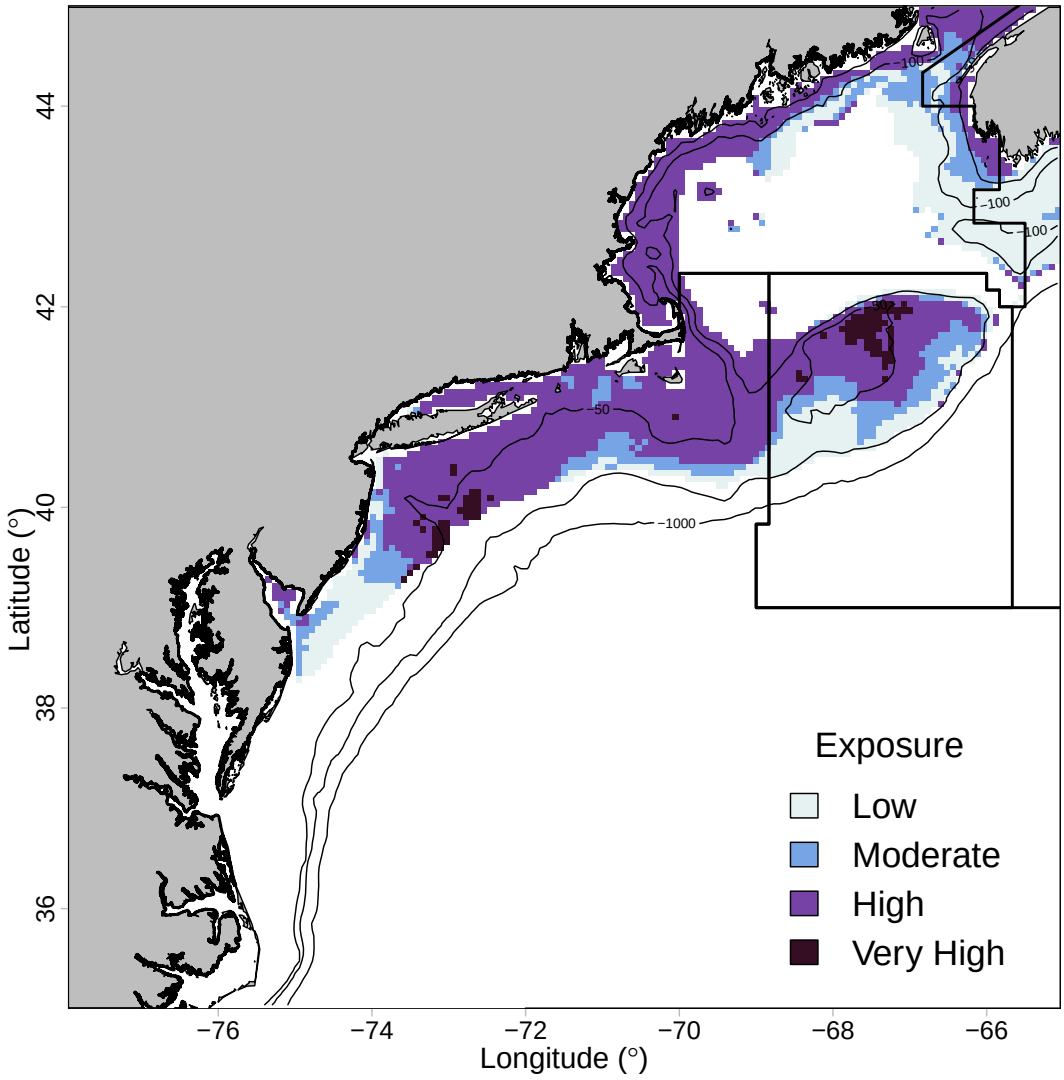


Figure 5: Total Exposure on the Northeast Shelf for Winter flounder. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 4)

Vulnerability:

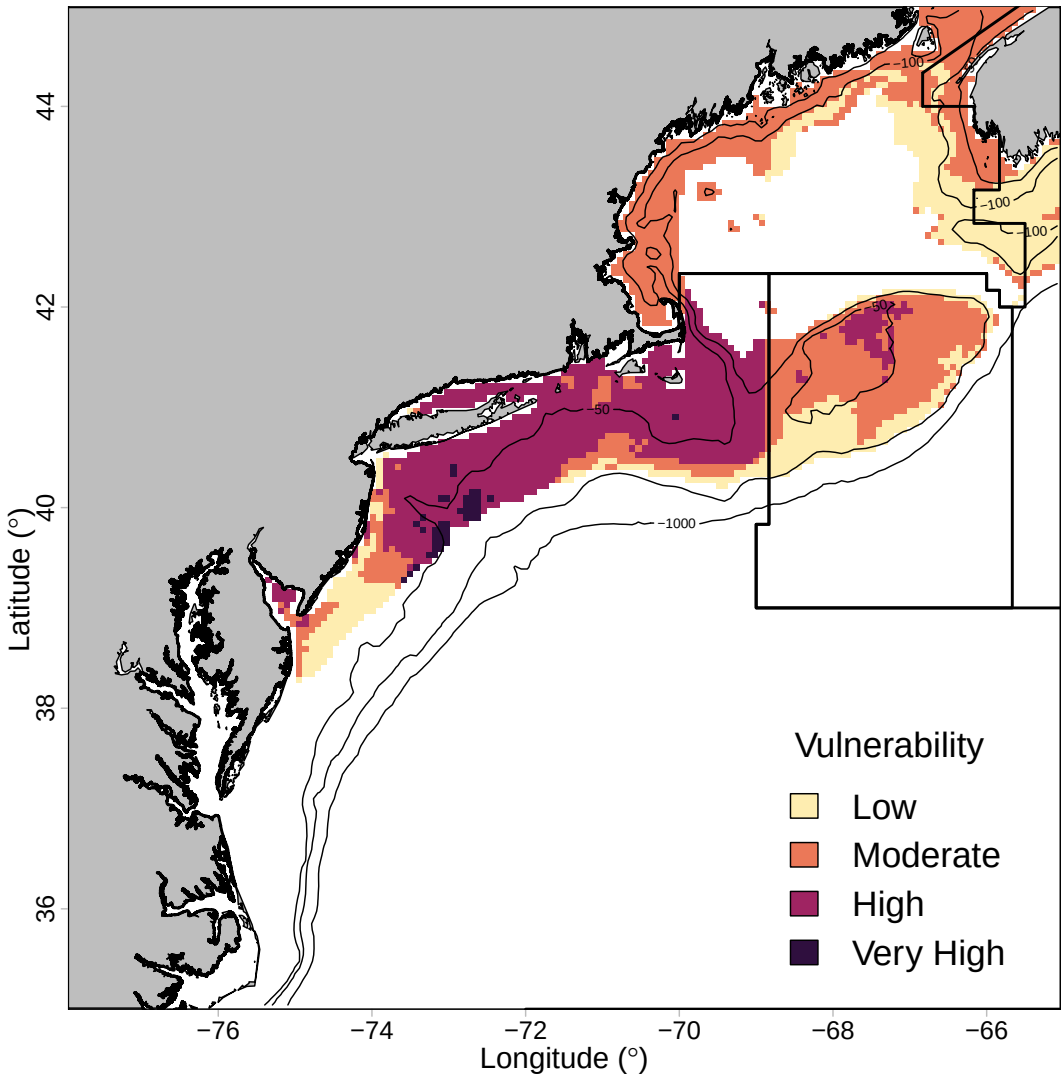


Figure 6: Vulnerability on the Northeast Shelf for Winter flounder. Vulnerability is the product of sensitivity and exposure (see Figure 5 and Figure 3)

Mean vulnerability across the range of winter flounder was **Moderate** (5.84 +/- 2.71) (Figure 1). Average scores were **Moderate** within the **Gulf of Maine** and **Georges Bank** stocks (4.41 +/- 1.79 and 4.58 +/- 1.95, respectively) and **High** (7.93 +/- 2.19)in the **Southern New England/Mid Atlantic** stock (Figure 1). Scores were High throughout Southern New England and over parts of Georges Bank and Moderate along the coast of the Gulf of Maine (Figure 6).

In the previous Assessment, winter flounder received a Very High vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

Between 95 and 100% of data quality scores were greater than or equal to 2 for both stocks of winter flounder, indicating excellent data quality. The only data gap (mean attribute data quality score < 2) identified was Stock Size/Status for the Gulf of Maine stock (Figure 1).

Model Confidence Score:

The winter flounder ensemble distribution model received a 2.5 (+/- 0.1) mean model confidence score, indicating high confidence in the final model. While the overall distributions matched experts' expectations, experts highlighted that the model missed known inshore and offshore migration patterns. The ensemble model AUC score was 0.92, indicating excellent model performance.

Additional Metrics

Distribution Shifts:

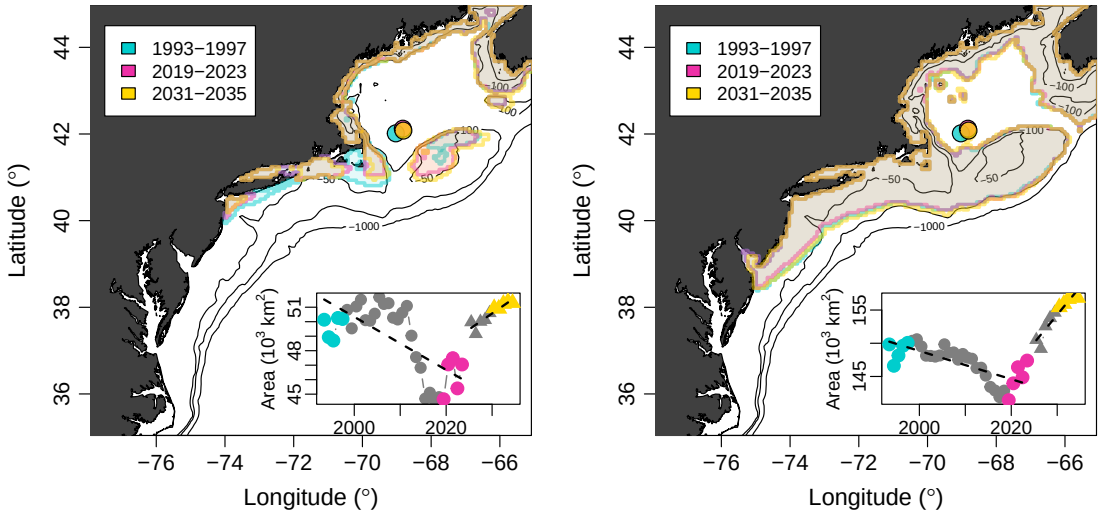


Figure 7: Distribution shifts on the Northeast Shelf for Winter flounder. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Winter flounder core habitat and overall range both declined significantly between 1993 and 2023. In both cases, habitat shifted towards more inshore habitats. Core habitat also expanded on Georges Bank (Figure 7). During this time, the weighted centroid shifted 21 km east-northeast.

Under near-future ocean conditions, winter flounder core habitat and range are predicted to expand significantly. Core habitat is predicted to increase offshore and on Georges Bank. Winter flounder range is expected to move offshore close to 1993 boundariesv(Figure 7). The weighted centroid is predicted to shift 5 km south-southeast.

In the previous assessment, winter flounder received a High Distributional Vulnerability Rank score, due to their varied habitat preferences and known migration patterns (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	49,744	149,101	4.74	8.10
2019-2023	20.7 km ENE	46,414 (-3.330)	144,972 (-4,129)	4.49 (-0.25)	7.94 (-0.16)
2031-2035	5.0 km SSE	51,137 (+4,723)	156,383 (+11,411)	4.61 (+0.12)	8.02 (+0.08)

Directional Effect:

Climate change was predicted to **negatively** impact all three stocks of winter flounder. Certainties ranged from 58% - 100% across stocks, with the lowest certainty for the Gulf of Maine stock and the highest for the Southern New England/Mid-Atlantic stock. Increasing temperatures have been linked to poor spring recruitment and higher summer time mortality, in addition to shifts in migration timing (Able et al. 2014) (Langan 2021) (Klein et al. 2017). The Southern New England/Mid Atlantic stock has also shifted norward (Able et al. 2014) (Langan 2021). The previous Assessment also predicted that winter flounder would be negatively impacted by climate change (Hare et al. 2016).

Climate Effects on Abundance and Distribution

A number of studies describe evidence of changes in winter flounder growth, abundance, and distribution as a result of climate change. The Southern New England/Mid-Atlantic Bight stock declined sharply in the 1980s from overfishing, has not recovered, and is shifting northward (Able et al. 2014) (Langan 2021). Warm spring temperatures have resulted in poor recruitment (Able et al. 2014), and declining productivity due to warming estuarine waters has blocked recovery efforts (Bell et al. 2014). Elevated water temperatures can cause faster development rates of early life history winter flounder; length of incubation time is crucial as a nearshore and estuarine species as these areas are more susceptible to anthropogenic impacts (Klein et al. 2017). Increasing temperatures may alter predation rates, affecting survival of larval winter flounder, as well as migration and distribution patterns (Klein et al. 2017). In Narragansett Bay, warmer summer temperatures were linked to increasing mortality in young-of-the-year winter flounder. Greater abundance of mature winter flounder in the lower bay during spring may also indicate a potential shift in spawning migration timing (Langan 2021).

Life History Synopsis

Winter flounder are found in estuaries and on the continental shelf from Nova Scotia to North Carolina, and are most common in the Mid-Atlantic north of Delaware Bay (Pereira 1999). They prefer temperatures between 0-20°C and depths less than 30m inshore and 100m offshore (Pereira 1999). The fishery is divided into 3 stocks: Gulf of Maine, Georges Bank, and Southern New England/Mid-Atlantic. They reach maturity at 2- 3 years, varying across stocks with southern populations maturing younger than northern ones, and live to a maximum age of 15 years (Northeast Fisheries Science Center (NEFSC) 2011). Winter flounder spawn inshore in the winter through spring, peaking in February/March in the Mid-Atlantic and Southern New England and continuing into May in the Gulf of Maine (Pereira 1999). Timing of spawning aggregations is linked to inshore migration timing, which is complex and possibly related to time of year, temperature, lunar cycle, and photoperiod, as well as varying across years. Individuals migrate inshore in the fall, and migrate offshore in the spring. The Georges Bank population does not migrate (Pearcy 1962). Females deposit eggs at night and on bottom substrates, usually sand or algal mats, and can deposit multiple batches of eggs per spawning season (Ziegler et al. 2019). Eggs are demersal and hatch within 2-3 weeks, depending on temperature. Larvae are initially planktonic but become more demersal as they metamorphosize, which starts 5-6 weeks after hatching. Larvae rely on bottom currents to keep them in and around spawning areas in inshore bays and estuaries. They are tolerant to a wide range of temperatures, salinities, and dissolved oxygen concentrations (Pearcy 1962). Metamorphosis is complete about 8 weeks after hatching, and is temperature dependent (Pereira 1999). Size, rather than age, tends to drive metamorphosis and settlement (Manderson et al. 2003). Larvae consume a variety of planktonic prey including nauplii, invertebrate eggs, calanoids, and polychaetes, with larger larvae consuming larger prey. Copepods and other zooplankton are important during metamorphosis (Stehlik and Meise 2000). Small medusae are important predators to larvae. Larvae can also be preyed upon by amphipods and Atlantic mackerel (Pereira 1999). Juveniles utilize marine and estuarine rocky, sand, and mud substrates, particularly fine sand and gravel as well as marine and estuarine water columns. They also occupy submerged aquatic vegetation, marshes, oyster reefs, and macroalgae habitats (Bigelow and Schroeder 1953). Juveniles consume a variety of prey including copepods, amphipods, and polychaetes. The diversity of prey increases and the importance of smaller zooplankton decreases with size (Pearcy 1962) (Pereira 1999). Adults are omnivorous opportunistic feeders. They feed during the day as they depend on sight to locate prey. Their diet is limited by mouth size; diets primarily consist of polychaetes and crustaceans (mostly amphipods), but also include bivalves, fish eggs, and fish. Diets can shift based on available prey (Stehlik and Meise 2000) (Pereira 1999). Predators of adult winter flounder include striped bass, spiny dogfish, and seals (Pereira 1999). The New England Fishery Management Council manages winter flounder in federal waters under the Northeast Multispecies (Groundfish) Fishery Management Plan. The Atlantic States Marine Fisheries Commission coordinates management in state waters. None of the 3 stocks are subject to overfishing. The Georges Bank and Southern New England/Mid-Atlantic stocks are not overfished; the Gulf of Maine stock has not been assessed and population abundance is unknown (Northeast Fisheries Science Center (NEFSC) 2025b) (Northeast Fisheries Science Center (NEFSC) 2025c) (Northeast Fisheries Science Center (NEFSC) 2025a).

Supplemental Scores

Exposure Timeseries

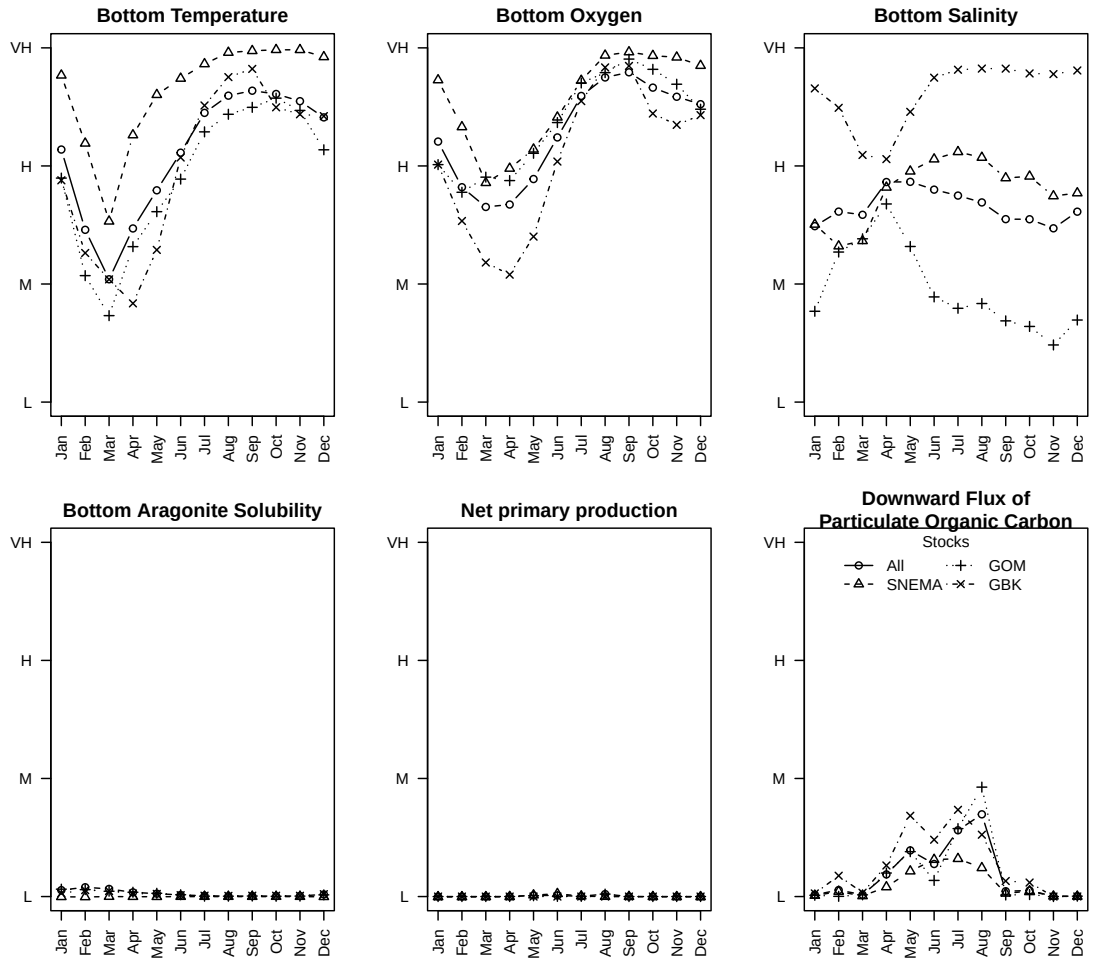


Figure 8: Monthly Timeseries of each Exposure Factor for Winter flounder. Exposure was calculated between 2014-2023 and 2025-2035

Exposure ranged from Moderate to High across the species range but there was higher variability across months within the different stock regions (Figure 9)). Georges Bank had the highest variability, with exposure ranging from Low in the spring (March - May) and Very High in late summer (July - September). In the Gulf of Maine stock, exposure was Low in February - March and High from July through December. The Southern New England/Mid Atlantic stock was the least variable, ranging from Moderate to High and following the same patterns as the entire species range (Figure 9)). These patterns were generally driven by trends in bottom temperature and bottom oxygen exposure. Bottom salinity exposure was greatest across months in the Georges Bank stock region, driving trends there (Figure 8)).

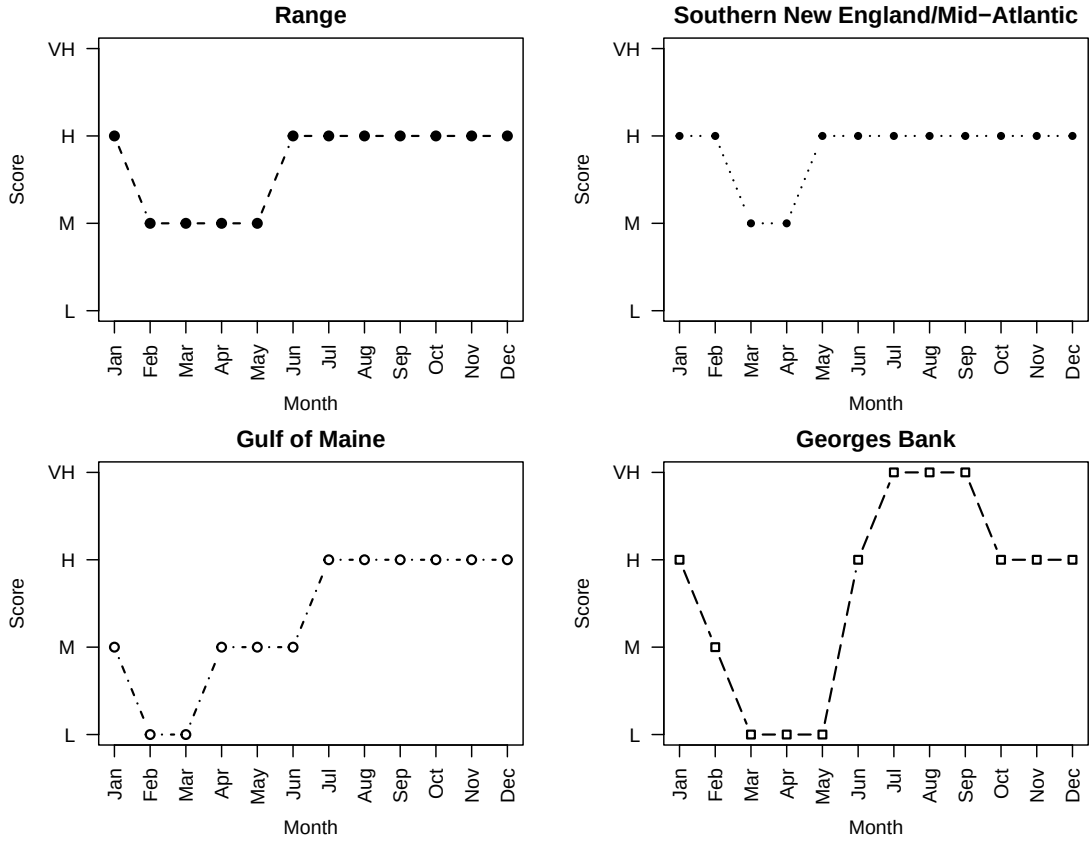


Figure 9: Timeseries of Total Exposure for Winter flounder

Exposure due to Significant Factors

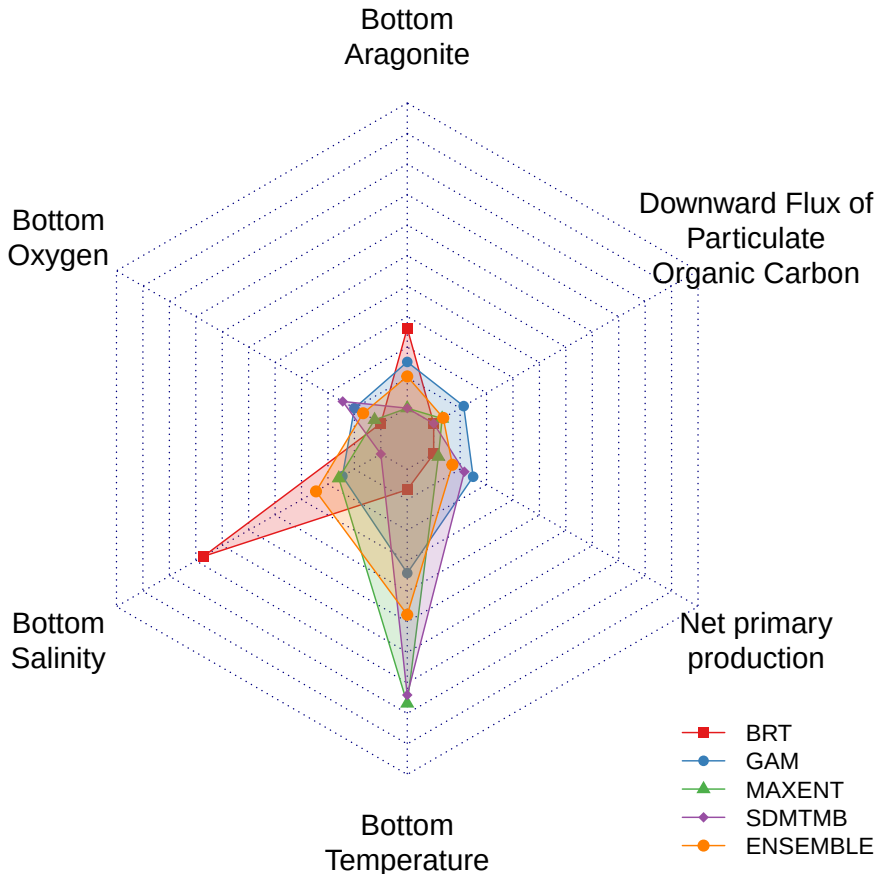


Figure 10: Dynamic Factors driving Winter flounder distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, salinity, and aragonite solubility were the major dynamic drivers of winter flounder distributions (Figure 10). Winter flounder are demersal and occupy the continental shelf and inshore habitats like estuaries (Pereira 1999). Shellfish beds can be important habitats, especially for juveniles (Bigelow and Schroeder 1953). Adults are opportunistic feeders and prey is mostly limited by mouth size (Stehlik and Meise 2000) (Pereira 1999). When only these factors were used to calculate exposure, scores mostly remained Moderate. Scores decreased from Moderate to Low in the Gulf of Maine stock (Figure 12, Figure 11).

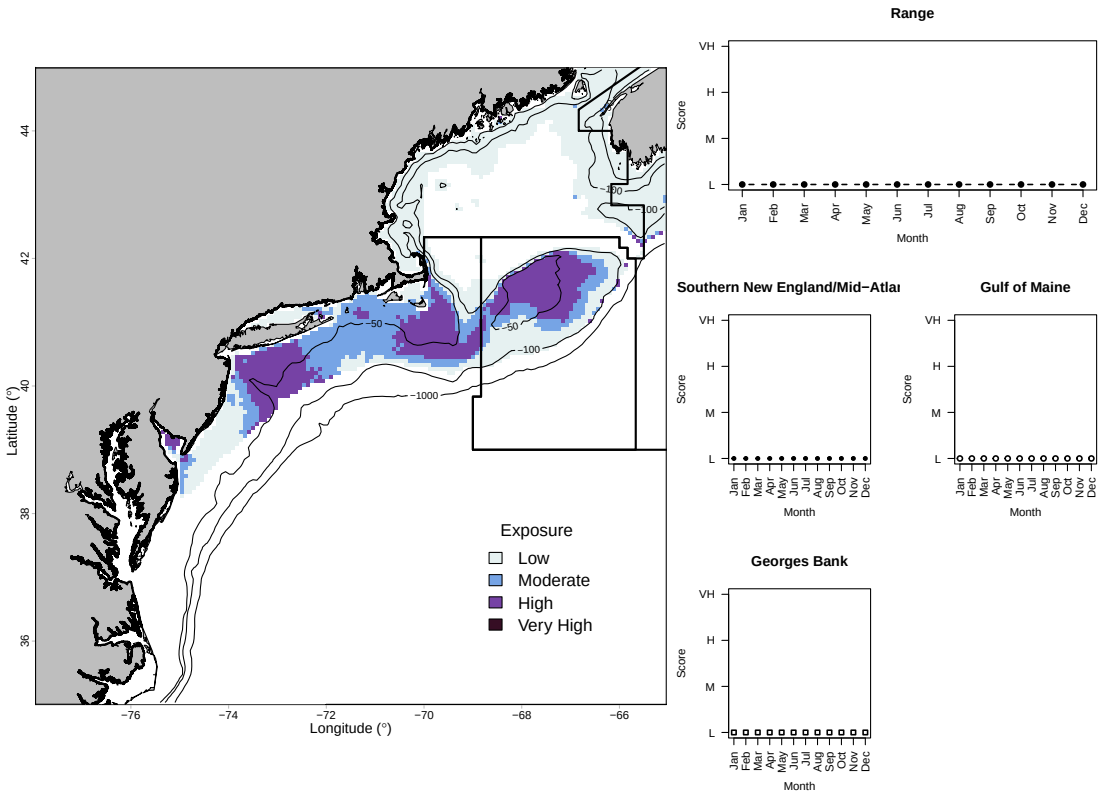


Figure 11: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Winter flounder distributions



Sensitivity

Sensitivity Attributes	Range Score	Range LH DQ	Range Tally	Gulf of Maine Score	Gulf of Maine LH DQ	Gulf of Maine Tally	Georges Bank Score	Georges Bank LH DQ	Georges Bank Tally	Southern New England/Mid-Atlantic Score	Southern New England/Mid-Atlantic LH DQ	Southern New England/Mid-Atlantic Tally
Adult Mobility	1.67	2.8		1.64	2.8		1.6	2.8		1.76	2.8	
Complexity in Reproductive Strategy	3.43	2.64		3.44	2.55		3.32	2.68		3.52	2.68	
Dispersal of Early Life Stages	2.37	2.8		2.44	2.7		2.28	2.8		2.4	2.9	
Egg and Larval Survival and Settlement Requirements	2.92	2.75		2.92	2.74		2.68	2.75		3.16	2.75	
Historical Variability in Temperature	2.03	2.87		2.2	2.9		1.96	2.8		1.92	2.9	
Juvenile Habitat Requirements	2.23	2.9		2.32	2.9		2.04	2.9		2.32	2.9	
Other Stressors	2.85	2.68		2.92	2.55		2.56	2.75		3.08	2.75	
Population Growth Rate	1.91	2.81		2	2.68		1.92	2.88		1.8	2.88	
Prey Specificity	1.57	3		1.56	3		1.52	3		1.64	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.24	2.9		1.24	2.9		1.16	2.9		1.32	2.9	
Spawning Cycle	2.84	2.77		2.84	2.7		2.84	2.8		2.84	2.8	
Stock Size/Status	2.65	2.53		2.72	1.9		2.96	2.9		2.28	2.8	
Total Sensitivity | Certainty	**Moderate (2) | 0.81**			**Moderate (2) | 0.58**			**Moderate (2) | 0.58**			**High (3) | 0.91**		

Exposure

Exposure Factors	Range Mean	Range SD	Range Tally	Gulf of Maine Mean	Gulf of Maine SD	Gulf of Maine Tally	Georges Bank Mean	Georges Bank SD	Georges Bank Tally	Southern New England/Mid-Atlantic Mean	Southern New England/Mid-Atlantic SD	Southern New England/Mid-Atlantic Tally
Bottom Temperature	2.92	0.98		2.65	1.07		2.7	0.86		3.42	0.65	
Bottom Salinity	2.76	0.89		2.09	0.89		3.5	0.63		2.83	0.61	
Bottom Aragonite Solubility	1.05	0.26		1.06	0.27		1.08	0.35		1	0.04	
Total Exposure	**Moderate (1.68)**	**0.83**		**Low (1.02)**	**0.16**		**Moderate (2)**	**0.89**		**Moderate (2.06)**	**0.80**	

Vulnerability

Vulnerability	Range Mean	Range SD	Range Tally	Gulf of Maine Mean	Gulf of Maine SD	Gulf of Maine Tally	Georges Bank Mean	Georges Bank SD	Georges Bank Tally	Southern New England/Mid-Atlantic Mean	Southern New England/Mid-Atlantic SD	Southern New England/Mid-Atlantic Tally
Total Vulnerability	**Moderate (4.22)**	**2.55**		**Low (2.03)**	**0.32**		**Moderate (4)**	**1.79**		**Moderate (6.14)**	**2.41**	

Low Moderate High Very High

Figure 12: As in Figure 1 for Winter flounder, but with only significant Exposure factors and the resulting Vulnerability Scores.

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